

encryption using Curve25519 for key exchange, establishing a tunnel nearly instantaneously, and is also significantly easier to configure and audit compared to previous generations.

OpenVPN is a well-tested and versatile VPN solution that can support both TLS-based tunnels and more platforms overall than WireGuard.

OpenSSH for administration and file transfer: OpenSSH is so ubiquitous that it is easy to forget it is an encryption tool. Every SSH session, SFTP transfer, and SCP copy is protected with strong modern ciphers, replacing the plaintext Telnet and FTP protocols of the past. SSH also provides secure port forwarding, allowing legacy applications that cannot speak TLS themselves to be tunnelled through an encrypted channel.

Encrypting data at rest with open source tools

Full-disk encryption — LUKS and dm-crypt: On Linux, the standard for block-device encryption is dm-

crypt, managed through the LUKS (Linux Unified Key Setup) format via the crypt setup utility. LUKS encrypts entire disks or partitions with AES-256 in XTS mode, so every file, temporary file, and swap page written to the device is automatically ciphertext. Its key-slot design allows up to eight independent passphrases or key files to unlock the same volume, enabling recovery keys and multi-administrator access. The ideal option for laptops, servers, and USB flash drives is full-disk encryption that serves the primary level of protection against the most common and basic types of viral threats. Many installers for Linux nowadays provide the capability to encrypt a hard disk using LUKS encryption via one box on the installation program. Many cloud VMs can use this method of encrypting attached volumes.

Cross-platform containers – VeraCrypt: TrueCrypt has been replaced by VeraCrypt, which is an audited successor that offers encrypted disks

and containers for Windows, MacOS X, and Linux systems. The main feature of VeraCrypt is that it offers a way to create an encrypted container (file). This means you can create a single file that can be mounted as a virtual disk to encrypt your folder of sensitive files and/or transfer a USB stick with encrypted data between multiple operating systems securely. VeraCrypt also supports hidden volumes which provide plausible deniability by hiding an encrypted volume in the free space of another encrypted volume.

File-level encryption: GnuPG and Age

Disk encryption secures data only when the drive is disconnected from a power supply or other devices, but when the operating system has been loaded and the hard disk has been unlocked, file data is no longer encrypted and any user or application that has the permission can obtain read access. File encryption adds a second finer level of protection.

Table 1: A comparison of open source encryption tools

Tool	Category	Encryption scope	Key features
OpenSSL	In transit	TLS/SSL for network traffic	Industry-standard TLS 1.3 library; certificate generation; wide protocol and cipher support
Let's Encrypt (Certbot)	In transit	HTTPS certificates	Free automated CA; 90-day auto-renewing certificates; ACME protocol
WireGuard	In transit	VPN tunnels	Modern, lightweight VPN; ChaCha20-Poly1305; built into the Linux kernel
OpenSSH	In transit	Remote access and file transfer	Encrypted shell sessions, SFTP/SCP, port forwarding and tunnelling
LUKS/dm-crypt	At rest	Full-disk/volume encryption	AES-256-XTS; native Linux block-device encryption; multiple key slots
VeraCrypt	At rest	Disk and container encryption	Cross-platform; hidden volumes; plausible deniability; TrueCrypt successor
GnuPG (GPG)	At rest	File and email encryption	OpenPGP standard; hybrid public-key encryption; digital signatures
Age	At rest	File encryption	Simple modern CLI; X25519 keys; minimal configuration
pgcrypto (PostgreSQL)	At rest	Column-level database encryption	In-database encryption functions for sensitive fields
HashiCorp Vault	Key management	Secrets and key lifecycle	Centralised secrets storage; dynamic credentials; encryption-as-a-service; audit logging